or the faster we move (chapter 3), the more time slows down: there is no single duration between two events; there are many possible ones.

The rhythms at which time flows are determined by the gravitational field, a real entity with its own dynamic that is described in the equations of Einstein. If we overlook quantum effects, time and space are aspects of a great jelly in which we are immersed (chapter 4).

But the world *is* a quantum one, and gelatinous spacetime is also an approximation. In the elementary grammar of the world, there is neither space nor time—only processes that transform physical quantities from one to another, from which it is possible to calculate probabilities and relations (chapter 5).

At the most fundamental level that we currently know of, therefore, there is little that resembles time as we experience it. There is no special variable "time," there is no difference between past and future, there is no spacetime (Part Two). We still know how to write equations that describe the world. In those equations, the variables evolve with respect to each other (chapter 8). It is not a "static" world, or a "block universe" where all change is illusory (chapter 7): on the contrary, ours is a world of events rather than of things (chapter 6).

This was the outward leg of the journey, toward a universe without time.

The return journey has been the attempt to understand how, from this world without time, it is possible for our perception of time to emerge (chapter 9). The surprise has been that, in the emergence of familiar aspects of time, we ourselves have had a role to play. From *our perspective*—the perspective of creatures who make up a small part of the world—we see that world flowing in time. Our interaction with the world is partial, which is why we see it in a blurred way. To this blurring is added quantum indeterminacy. The ignorance that follows from this determines the existence of a particular variable—thermal time (chapter 9)—and of an entropy that quantifies our uncertainty.

Perhaps we belong to a particular subset of the world that interacts with the rest of it in such a way that this entropy is lower in one direction of our thermal time. The directionality of time is therefore real but perspectival (chapter 10): the entropy of the world *in relation to us* increases with our thermal time. We see the occurrence of things ordered in this variable, which we simply call "time," and the growth of entropy distinguishes the past from the future for us and leads to the unfolding of the cosmos. It determines the existence of traces, residues, and memories of the past (chapter 11). We human beings are an effect of this great history of the increase of entropy, held together by the memory that is enabled by these traces. Each one of us is